

Investigation of a Pitch Function for Motion-Based VR Bicycle Simulators

Yu Wang
TU Wien
Vienna, Austria
yu.wang@tuwien.ac.at

Martin Steinmetz
TU Wien
Vienna, Austria
m.steinmetz@hotmail.de

Sonja Dorfbauer
TU Wien
Vienna, Austria
sonja.dorfbauer@tuwien.ac.at

Florian Michahelles
TU Wien
Vienna, Austria
florian.michahelles@tuwien.ac.at

Philipp Wintersberger
Univ. of Applied Sciences Hagenberg
Hagenberg, Austria
philipp.wintersberger@fh-hagenberg.at



Figure 1: The bike simulator consists of a city bike (1) mounted on a board (2), a bike trainer (3), the VR headset (4), an airstream simulation (5), the motion platform (6), and sensors for brakes and steering connected to an Arduino (7). In our study, participants had to cycle a track with flat, up-, and downhill sections while being supported by different pitch modes.

ABSTRACT

Research on cycling safety and comfort is becoming increasingly relevant in the domain of human-computer interaction. Many studies presented so far have been conducted on static bicycle simulators, which suffer problems of immersion and simulator sickness. Since previous publications on simulator design have suggested a positive effect of simulator tilt (lateral motion), we investigated the potential of simulator pitch (longitudinal motion). N=31 participants completed a test track with multiple flat, up-, and downhill sections while being supported with different pitch modes and a baseline without any motion. Our results suggest only a small impact of simulator pitch on simulator sickness and perceived immersion. However, we found out that study participants adjusted their pedaling strength based on the inclination of the test track, which we

attribute to the plausibility illusion in virtual reality. The experiment contributes to the design of virtual reality bicycle simulators at different degrees of fidelity.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI.

KEYWORDS

Cycling simulation, physical simulator, virtual reality, simulator sickness, presence, immersion

ACM Reference Format:

Yu Wang, Martin Steinmetz, Sonja Dorfbauer, Florian Michahelles, and Philipp Wintersberger. 2023. Investigation of a Pitch Function for Motion-Based VR Bicycle Simulators. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585623>

1 INTRODUCTION

Bicycle simulators are an increasingly important tool in human-computer interaction research. Since cycling is an environmentally-friendly and healthy transportation modality that may act as an

alternative to traditional automobiles in the future, more and more scientific papers address cyclists' safety and comfort [2, 25–28, 48]. Bicycle simulators play an important role in this development, as they (just like simulators in other transport-related settings) allow the investigation of potentially dangerous situations in relatively safe laboratory environments. However, while car/vehicle, flight, or motorcycle simulators have become sophisticated high-technology products over the years, literature on the effects of bicycle simulators on user perception is still scarce [53]. There exist commercially available products to bike in virtual environments (such as "Zwift"), but these systems focus on fitness and entertainment rather than a realistic replication of cycling. So far, most HCI-related studies utilized static bicycle simulators, which do not necessarily feel realistic, can result in motion sickness, and increase the dropout rates among study participants [6, 7, 49]. Motion/Simulator sickness usually arises from a sensory conflict between the visual, vestibular, and/or proprioceptive systems [17]. Since static setups cannot accurately simulate many bicycle dynamics, they are limited by default – especially when users steer into curves. A recent study has suggested that motion sickness symptoms can partly be reduced with minor modifications such as an appropriate steering method or the provision of an airstream simulation [29]. However, one typical way to mitigate motion sickness and increase the realism (i.e., "fidelity") of simulators is the inclusion of physical motion. Regarding motion/simulator sickness and experienced realism, motion-based simulators have demonstrated their benefits over static simulations in the context of aviation [18] and vehicles [3]. Further, a literature review by [5] suggests that higher simulator fidelity (in particular, visual fidelity) positively affects simulator sickness and that this is "primarily true for dynamic motion simulators".

Regarding bicycle simulators, a realistic perception of motion could be highly relevant. According to Kooijman and Schwab [20], cyclists do not only control the bike, they "contribute significantly to the mechanical system", and a rider's weight accounts for up to 90% of the complete system's mass. We argue that bicycle simulators should aim at reproducing the complex motion dynamics on real bicycles to minimize sensory mismatch and provide a realistic cycling experience. Consequently, we built a motion-based virtual reality (VR) bicycle simulator that allows lateral and longitudinal motion. In a recent publication, we could show that a properly applied tilt function (i.e., lateral simulator motion) can significantly increase presence/immersion without raising simulator sickness [53]. Given the demonstrated benefits of including a simulator tilt for cycling in VR, it could also be interesting to evaluate the effects of simulator pitch. Longitudinal motion is considered an important feature of driving simulators to communicate acceleration and forces to users [1]. Pitch functions have been investigated with motorcycle simulators, which can be considered a "near relative" of bicycle simulators. Thereby, the effects of pitch on simulator sickness are not completely clear, but it was suggested that pitch functions can lead to more valid braking velocity and higher experienced realism [10]. As we are not aware of studies specifically addressing pitch for bicycle simulators, we conducted a user study with different pitch modes, where participants had to cycle a test track with multiple up- and downhill sections as well as flat and bumpy roads. While our results suggest that simulator pitch has only little effect on simulator sickness and perceived realism, we revealed some

interesting results regarding participants' pedaling behavior. In particular, cycling speed was significantly influenced by the visual perception of the environment, independent of physical motion. Ultimately, the results of this initial experiment motivate future research in this area.

2 RELATED WORK

Immersive simulators have been developed for entertainment (like skiing [31], skydiving [8], or climbing [41]) but also for tasks such as aviation, driving, or space travel. In safety-critical scenarios, simulators allow experimentation and training with minimal risk. Developing sophisticated bicycle simulators is challenging, since on a bicycle, human and machine are directly and closely coupled, and the riders' behaviors and movements significantly contribute to the overall dynamics [20]. Simulators can be evaluated in terms of objective and subjective validity. Objective validity refers to the capability of a simulator to accurately reproduce realistic outcomes [36]. In terms of subjective validity, both positive (i.e., immersion and perceived realism) and negative (i.e., simulator sickness) effects can influence users. Slater [40] has defined immersion as "*the extent to which a VR system can support natural sensorimotor contingencies for perception*". Presence, in turn, describes a user's perception of actually being in that environment [38]. Slater [39] further argues that successful VR environments create so-called "place illusions" and "plausibility illusions". While place illusions relate to presence, plausibility illusions describe "*the overall credibility of the scenario being depicted in comparisons with expectations*". Thereby, a conflict between expected and observed signals can lead to simulator sickness (also called motion- or cybersickness) [50]. In this context, Weech et al. [50] have argued that presence and simulator sickness are negatively related. Simulator sickness can be evaluated from three different dimensions: nausea, general discomfort/disorientation, and oculomotor dysfunction [16], which also constitute the primary evaluation degrees in the simulator sickness questionnaire (SSQ). Kennedy and his colleague [15] also indicated that in different categories of visually induced motion sickness (VIMS), there is always a difference in the distributions of the three dimensions.

To mitigate simulator sickness, hardware, visual, and visuo-vestibular modifications have been proposed to improve bicycle simulators [53]. Visio-vestibular modifications are based on physical stimulation of the vestibular system, such as galvanic feedback, airflow, vibration, etc. [7, 11, 12, 22, 24, 32, 43, 51, 52]. In contrast, visual modifications include blurring, vignette or blink [9, 35, 47]. Further, simulator sickness can be influenced by the display choice, and users react differently to screens, VR, cave environments, or auditory displays [14, 30, 46]. For example, a large display typically leads to less sickness than VR goggles, at the cost of immersion [30]. Finally, simulator sickness and immersion also depend on the chosen hardware. Simulators are typically categorized according to different "levels of fidelity" [53], where visual fidelity refers to the realism of the computer-generated environment, while physical fidelity influences users' perception of presence [13]. In the context of bicycles, low-fidelity simulators are stationary setups or ergometers, where steering is supported by potentiometers or attached VR controllers [4, 42, 46]. According to Matvienko et al.

[29], the choice of input controls (i.e., handlebar rotation vs. leaning into the desired direction) influences perceptions, too.

Motion-based simulators are typically attributed to higher fidelity, and the positive influence of lateral motion (i.e., tilting of the bike) has been demonstrated [53]. Longitudinal motion is featured by most motion-based simulators. For flight simulators, Kim et al. [19] could show that cue mismatches in pitch angles correlate with motion sickness. However, most existing works evaluate the full set of dynamic features to static simulations [3, 45] without focusing on the longitudinal axis. In the context of motorcycle simulators, the findings seem to be inconsistent. While a study by Kováčová et al. [21] suggests that motion can increase workload without benefiting simulator sickness and braking behavior, Grotto et al. [10] identified a positive effect on braking velocity and presence. Still, longitudinal motion is included in most motion-based vehicle simulators to reproduce forces associated with acceleration. Consequently, we aimed at evaluating if longitudinal motion dynamics should be considered in the development of future VR bicycle simulators, too. Thereby, we investigated the effects of uneven surfaces (i.e., cycling on flat roads, up-, and downhill) in combination with different pitch modes of a VR bicycle simulator.

3 APPARATUS

The bicycle simulator used for this experiment consists of a NextLevel V3 Motion Platform that supports up to 10° rotation in both the longitudinal and lateral axis. Further, the simulator can simulate airstream and has sensors to measure the steering angle and braking behavior (see Figure 1). For more details on the technical setup, we refer to [53]. For this study, we did not implement any tilt as we solely focus on the longitudinal motion dynamics. We implemented two pitch modes (Elevation Mode, Full Mode) and a non-pitch mode (Baseline) as described in the following:

- **Elevation Mode:** This pitch mode reflects the elevation profile of the test track, i.e., up- and downhill gradients are mirrored on the longitudinal axis one-to-one. This mode guarantees a coupling between the visual and physical impressions of cycling flat, up-, and downhill.
- **Full Mode:** This mode is based on the *Elevation Mode* but additionally includes the forces given by acceleration and deceleration, i.e., strong pedaling (backward pitch) and braking (forward pitch) are added to the pitch provided by the elevation profile. The pitch angles were calculated based on gravity and speed differences, and the strength of the additional pitch was adjusted based on pilot tests with lab members to provide a plausible experience. In contrast to the *elevation mode*, this implementation adds users' expectations of acceleration/deceleration with the physical response.
- **Baseline Mode:** In the baseline mode, all motion features of the platform are deactivated, i.e., neither acceleration nor elevation profile is reflected. This mode corresponds to a static simulator without a motion platform.

Since the platform does only allow pitch up to 10 degrees, higher values were cut at this point. Another relevant aspect of cycling on uneven surfaces is the physical resistance of pedaling. Cycling uphill increases the resistance as more force is needed to fight gravity.

In turn, cycling downhill may not require pedaling at all. Bike trainers such as the one used for developing the used simulators (Tacx Flux 2) support modifying the resistance to make pedaling harder for the user. However, they typically do not support a realistic implementation of cycling downhill, as this would mean that a motor in the trainer would have to increase speed automatically, without the need for a user to pedal. Given this limitation, we choose to not modify the physical resistance in our experiment at all. In other words, study participants always felt the same pedal resistance no matter the inclination of the test track. This limitation may negatively influence immersion but in turn, allows us to investigate the effect of simulated inclination on users' behavior. In particular, we asked ourselves if participants will adapt their pedaling behavior to adjust the bicycle speed to their expectations (i.e., produce a plausibility illusion and cycle faster when riding downhill).

4 HYPOTHESES

Based on the considerations (potential effect of physical motion on simulator sickness, immersion, and perceived realism) and limitations (lacking possibility to accurately simulate downhill rolling) discussed above, we proposed the following three hypotheses:

- **H1:** A pitch mode for a VR bicycle simulator can reduce motion sickness.
- **H2:** A pitch mode for a VR bicycle simulator can increase the immersion when cycling in VR.
- **H3:** The perception of inclination based on a VR bicycle simulator can affect cycling speed.

5 USER STUDY

We developed a VR environment with a test track in Unity3D consisting of a (straight) road following multiple flat, up-, and downhill sections (see Figure 1). The maximum inclination in the individual road segments varied between 5 and 15 degrees (mean inclination: 6 degrees), and the track was 650 meters long. The cycling duration for each condition (Elevation, Full, and Baseline Mode) was approximately 3 minutes. Participants completed the test drive in each condition (within-subjects design). The data on cycling speed was logged from the simulator software, and the data of participants' simulator sickness, presence, and immersion in VR were collected in post-condition questionnaires.

5.1 Participants and Procedure

A sample of N=31 participants (9 female, 22 male; M=25.7, SD=2.1 years) voluntarily completed the experiment. They rated their cycling habits as diverse (from n=7 only exceptionally to n=6 at least once per week). First, they completed a safety briefing, provided demographics, and performed a short test drive on the simulator. Then, participants completed test drives in each of the provided pitch modes. The order of conditions was varied in a way that half of the participants completed the baseline drive first, the other half at the end, and the pitch modes (*Elevation Mode* and *Full Mode*) were counterbalanced. After each condition, participants completed a post-condition survey. Finally, we conducted a small semi-structured interview with each participant. In total, the study lasted for about half an hour per subject.

5.2 Measurements

To investigate the effect of the pitch modes and elevation profiles, we combined self-ratings with driving performance measurements. To assess simulator sickness, we included the simulator sickness questionnaire (SSQ) that measures the dimensions of nausea, oculomotor, and disorientation with 16 items on a 4-point Likert scale. For measuring presence and immersion in the virtual environment, we utilized the Ingroup Presence Questionnaire (IPQ). The IPQ consists of 14 items that measure the dimensions of spatial presence, involvement, and perceived realism (assessed on a 7-point Likert scale). Additionally, we measured participants' cycling speed every 100 milliseconds and averaged the resulting values for the three inclination modes *Uphill* (>1 degree), *Flat* (from 1 to -1 degrees), and *Downhill* (<-1 degree), per participant.

5.3 Results

IBM SPSS V24 was used to evaluate the data. As all scale dimensions yielded acceptable reliability (Cronbach's alpha > .6) we calculated scale values. Since the data did not follow a normal distribution, we conducted non-parametric Friedman tests with subsequent pairwise comparisons (Wilcoxon, lowered significance threshold of $p < .016$ to account for multiple post-hoc tests). Descriptive statistics are visible in Table 1.

5.3.1 Simulator Sickness. We calculated the simulator sickness total score and the three dimensions nausea, oculomotor, and disorientation. Since tests for normal distribution failed, we performed non-parametric Friedman tests. We found a significant effect for the dimension nausea ($\chi^2(2) = 29.069, p < .001$, see Figure 2 Post-hoc comparisons (Wilcoxon) revealed that the *Full Mode* led to significantly less nausea among study participants than the *baseline* ($Z = -4.199, p < .001$) and the *Elevation Mode* ($Z = -3.829, p < .001$), while there was no difference between the latter two. However, we did not find any significant effects for the other two dimensions and the SSQ total score.

5.3.2 Presence and Perceived Realism. Given that the dimensions of the IPQ scale (spatial presence, involvement, perceived realism) followed a normal distribution, we performed a series of repeated measures ANOVAs. However, we could not find any differences for the dimensions of the IPQ scale. In other words, the three conditions were perceived as similarly realistic considering the subjective ratings of our study participants.

5.3.3 Driving Performance. We compared participants' cycling speeds based on the three conditions and the different sections of each trip (*Uphill*, *Downhill*, and *Flat*). While there was no difference indicated between the pitch modes, the speed was significantly affected by the elevation profile in all conditions. In condition *Full Mode* ($\chi^2(2) = 41.097, p < .001$), participants pedaled significantly faster when cycling *uphill*, compared to both cycling *downhill* ($Z = -4.821, p < .001$) and *flat* ($Z = -4.546, p < .001$). Further, the speed differed significantly between *flat* and *downhill* sections ($Z = -2.508, p = .012$). The same result was present in the *Elevation Mode* condition ($\chi^2(2) = 41.097, p < .001$). The cycling speed was faster *uphill* than *downhill* ($Z = -4.840, p < .001$) and *flat* ($Z = -4.664, p < .001$), as well as *downhill* than *flat* ($Z = -2.548, p = .011$). Finally, the effect was also visible in the

Baseline condition ($\chi^2(2) = 42.065, p < .001$). Again, participants cycled significantly faster *uphill* than *flat* ($Z = -4.586, p < .001$) and *downhill* ($Z = -4.625, p < .001$), and the speed downhill was significantly faster than *flat* ($Z = -2.646, p < .001$).

6 DISCUSSION

In this study, we implemented different pitch modes on a VR bicycle simulator to investigate if the higher fidelity of longitudinal simulator motion (i.e., pitch) influences simulator sickness, perceived realism, and cycling speed.

6.1 Pitch Modes

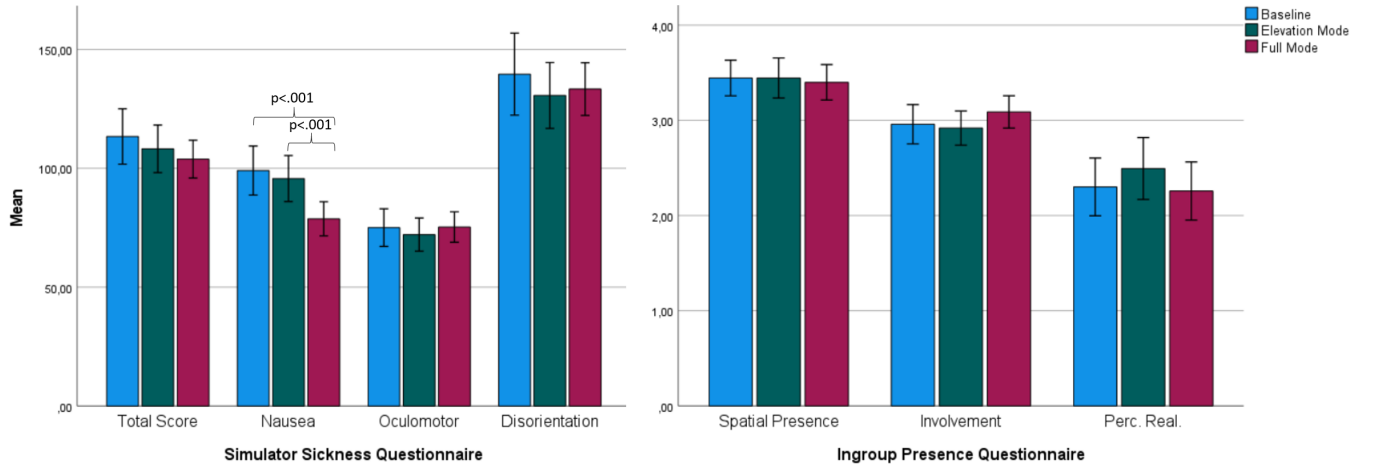
Our results do not suggest a strong influence of longitudinal simulator motion on simulator sickness. We accept **H1** as the nausea dimension of the SSQ was significantly affected, where the *Full Mode* yielded lower nausea than the other dimensions. The other two SSQ dimensions and the total SSQ score did not differ. Also, there was no influence of the pitch mode on immersion as assessed by the IPQ scale. Participants rated the dimensions of spatial presence, involvement, and perceived realism similarly in all three conditions. Consequently, **H2** must be rejected. This contrasts our previous study addressing simulator tilt, where we found effects on perceived realism but no influence on simulator sickness [53]. The obtained results are also contradictory to studies conducted on motorcycle simulators. In a study conducted by Grotoli et al. [10], motion did not affect simulator sickness but positively influenced participants' sense of presence. Also, Kováčová et al. [21] did not find an influence of motion on simulator sickness. Our findings are similar to experiments conducted in flight [19] or vehicle simulators [3]. Still, bicycle simulators are most similar to motorcycle simulators, and additional investigations are needed to draw reliable conclusions. Regarding perceived realism, there could be different reasons why simulator pitch (longitudinal motion) does not seem to influence immersion as simulator tilt (lateral motion), which has shown to be beneficial for VR cycling experiences [53]. One could be that tilting is just more relevant for matching users' expectations with bodily perceptions. Another reason could be the implementation of the pitch mode, which we implemented mainly based on lab members' impressions during a small pilot. Also, our bike trainer does not have a motor to allow automatically increasing the bike's speed when riding downhill. Consequently, we did not modify the pedaling resistance in our experiment. Thus, another reason could be that the combination of visual perception, pitch, and resistance did not match the study participants' expectations.

6.2 The Power of Virtual Reality for Plausibility and Coherence

The downside of not being able to modify pedaling resistance allowed us to investigate if users behave differently on the simulator when cycling flat, up-, and downhill. Here, we could see significant differences in cycling speed (**H3**). Participants pedaled fastest when cycling uphill, despite there was no physical resistance that would require them to do so. The result makes sense, as climbing uphill requires a cyclist to fight gravity and put more force into the longitudinal motion to maintain speed. Interestingly, we could see this effect in all conditions – even in the baseline, where no simulator

Table 1: Descriptive statistics (mean, standard deviation, and median) of the investigated experimental conditions Baseline Mode, Elevation Mode, and Full Mode

	Baseline Mean (SD, Med)	Elevation Mode Mean (SD, Med)	Full Mode Mean (SD, Med)
Simulator Sickness Questionnaire (SSQ)			
Total Score	113.4 (31.7, 112.2)	108.2 (27.3, 97.2)	103.9 (21.7, 101.0)
Nausea	99.0 (28.2, 85.8)	95.7 (26.4, 85.9)	78.8 (19.5, 76.3)
Oculomotor	75.1 (21.6, 68.2)	72.1 (19.1, 68.2)	75.3 (17.5, 68.2)
Disorientation	139.6 (47.1, 139.2)	130.7 (37.8, 111.4)	133.4 (30.2, 125.3)
Ingroup Presence Questionnaire (IPQ)			
Spatial Presence	3.44 (.51, 3.60)	3.45 (.57, 3.60)	3.40 (.51, 3.60)
Involvement	2.96 (.56, 2.75)	2.92 (.49, 3.00)	3.09 (.46, 3.00)
Percieved Realism	2.30 (.83, 2.33)	2.49 (.89, 2.33)	2.26 (.83, 2.33)
Cycling Speed			
Uphill	15.97 (3.06, 16.14)	15.29 (3.58, 15.66)	15.64 (3.59, 14.49)
Flat	13.11 (3.98, 12.26)	13.43 (4.50, 12.04)	12.70 (4.90, 10.37)
Downhill	14.29 (2.19, 14.63)	14.21 (3.16, 13.53)	13.80 (3.26, 13.31)

**Figure 2: Bar charts of participants' ratings (including 95% confidence interval) in the individual sub-scales of the simulator sickness questionnaire (SSQ) and the Ingroup presence questionnaire (IPQ). Only the nausea dimension of the SSQ significantly differed between the experimental conditions.**

pitch was present. This demonstrates that users' behavior in VR can be modified by visual stimulation only. We attribute this effect to the concepts of plausibility or coherence [23, 37, 39] since our participants behaved in a way to bring the credibility of the scenario in accordance with their own expectations. It is well known that humans' can be "tricked" in virtual environments due to the complex interplay of visual and bodily sensations. For example, approaches to simulate walking in restricted spaces like walking-in-place or redirected walking can still feel plausible [33, 34, 44]. Rietzler et al. [34] have demonstrated that a driving simulator can use rotation to simulate forward motion, arguing that "*different forms of non-matching stimuli should be tested*". In this context, the presented study has shown the power of illusions on user behavior when cycling in VR. Such effects may inform future (bicycle) simulator

development and suggest investigating simulator designs that do not necessarily foster highly realistic motion fidelity.

6.3 Limitations and Future Work

Our experiment shows various limitations which should be addressed in future work. First, the pitch function must be considered an initial design that was configured based on the perception of a small sample of lab members. Also, we used pressure sensors for braking instead of a real bicycle brake. Thus, the pitch resulting from braking was not necessarily physically accurate. We will improve the simulator design and investigate other pitch modes (i.e., using modifications like damping, scaling, or different onsets and cutoff values) systematically in the future. Regarding simulator sickness, the experiment lasted only three minutes per condition. Related work indicated that initial peaks in simulator sickness can

be reached early (i.e., within 5 minutes), but also that the severity increases over longer time periods [6]. Potentially, longer exposure could lead to the hypothesized differences between the conditions. As this is pure speculation, we will perform additional experiments with longer exposure times. Further, the used bike trainer does not allow an acceleration of the bike when rolling downhill, which could have influenced the results of the study, both in terms of simulator sickness and perceived realism. The design choice not to modify bike trainer resistance (neither in up- or downhill sections), in turn, allowed us to investigate the effect of possibility and coherence. In this context, future studies should investigate if these illusions would preserve over longer exposure times, but also if modifying the bike speed and resistance would influence these results. Ultimately, we call for more research on VR bicycle simulator fidelity in general. In particular, we aim to compare and investigate low-fidelity approaches such as VR cycling with a gamepad or handlebar only (i.e., without pedaling). Finally, involving more measures of human body motion and gesture, for instance, shoulder checks, head checks, etc. can also support the investigation of human cycling behavior in the future.

7 CONCLUSION

In this work, we have investigated if a physical simulator pitch supported by a motion platform can improve motion sickness and immersion in VR cycling. We implemented two different pitch modes based on the elevation profile of the test track and pitch according to acceleration/deceleration and conducted a user study with $N=31$ participants. Participants had to complete a test track with multiple flat, up-, and downhill sections in each pitch mode as well as a (static) baseline without any simulator motion and rate their impressions using the simulator sickness questionnaire and the ingroup presence questionnaire. Additionally, we recorded participants' cycling speeds in all sections of the test track. With the exception of the nausea component of simulator sickness, our results suggest that simulator pitch plays only a minor role for VR bicycle simulators, as the different conditions were rated similarly in all conditions. Overall, our study does not suggest that future VR bicycle simulators must provide longitudinal motion to raise fidelity. However, we could find an interesting result regarding the "plausibility illusion" in VR. Participants varied their pedaling strength based on the inclination of the test track to bring their expectations (cycling uphill is harder than flat or downhill) in accordance with their behavior. This effect was visible in all conditions, even in the static baseline where no pitch was present. We conclude by arguing that further investigation of such effects could be beneficial for the design of low-fidelity simulations – not only in the context of bicycles but simulator development in general.

ACKNOWLEDGMENTS

This publication is supported by the "Mobility of the Future" program of the Austrian Research Promotion Agency (FFG) under Grant No. FO999893961 (Eternity Bike).

REFERENCES

- [1] Imad Al Qaisi, Daniel Zimmermann, Andreas Kohlstedt, Sandra Gausemeier, and Ansgar Trächtler. 2014. Subjective evaluation of different motion cueing algorithms implemented on atmos driving simulator. In *Driving Simulation Conference* 2014. Driving Simulation Conference, Paris, France.

- [2] Josh Andres, Julian de Hoog, and Florian 'Floyd' Mueller. 2018. "I Had Super-Powers When EBike Riding" Towards Understanding the Design of Integrated Exertion. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play* (Melbourne, VIC, Australia) (CHI PLAY '18). Association for Computing Machinery, New York, NY, USA, 19–31. <https://doi.org/10.1145/3242671.3242688>
- [3] Baris Aykent, Frederic Merienne, Christophe Guillet, Damien Paillot, and Andras Kemeny. 2014. Motion sickness evaluation and comparison for a static driving simulator and a dynamic driving simulator. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering* 228, 7 (2014), 818–829.
- [4] Shani Batcir, Omri Lubovsky, Yaacov G. Bachner, and Itzhak Melzer. 2021. The Effects of Bicycle Simulator Training on Anticipatory and Compensatory Postural Control in Older Adults: Study Protocol for a Single-Blind Randomized Controlled Trial. *Frontiers in Neurology* 11 (2021). <https://doi.org/10.3389/fneur.2020.614664>
- [5] Ksander N de Winkel, Tessa MW Talsma, and Riender Happee. 2022. A meta-analysis of simulator sickness as a function of simulator fidelity. , 17 pages.
- [6] Natalia Duzmańska, Paweł Strojny, and Agnieszka Strojny. 2018. Can Simulator Sickness Be Avoided? A Review on Temporal Aspects of Simulator Sickness. *Frontiers in Psychology* 9 (2018). <https://doi.org/10.3389/fpsyg.2018.02132>
- [7] Sarah D'Amour, Jelte E. Bos, and Behrang Keshavarz. 2017. The efficacy of airflow and seat vibration on reducing visually induced motion sickness. *Experimental Brain Research* 235, 9 (Sept. 2017), 2811–2820. <https://doi.org/10.1007/s00221-017-5009-1>
- [8] Horst Eidenberger and Annette Mossel. 2015. Indoor Skydiving in Immersive Virtual Reality with Embedded Storytelling. In *Proceedings of the 21st ACM Symposium on Virtual Reality Software and Technology* (Beijing, China) (VRST '15). Association for Computing Machinery, New York, NY, USA, 9–12. <https://doi.org/10.1145/2821592.2821612>
- [9] Sebastian Freitag, Dominik Rausch, and Torsten Kuhlen. 2014. Reorientation in virtual environments using interactive portals. In *2014 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, MN, USA, 119–122. <https://doi.org/10.1109/3DUI.2014.6798852>
- [10] Marco Grottolli, Max Mulder, and Riender Happee. 0. Motorcycle simulator subjective and objective validation for low speed maneuvering. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering* 0, 0 (0), 09544070221110930. <https://doi.org/10.1177/09544070221110930> arXiv:<https://doi.org/10.1177/09544070221110930>
- [11] Germán Gálvez-García, Marion Hay, and Catherine Gabaude. 2015. Alleviating Simulator Sickness with Galvanic Cutaneous Stimulation. *Human Factors* 57, 4 (2015), 649–657. <https://doi.org/10.1177/0018720814554948>
- [12] Jake Harrington, Benjamin Williams, and Christopher Headleand. 2019. A Somatic Approach to Combating Cybersickness Utilising Airflow Feedback. *Computer Graphics and Visual Computing (CGVC)* -, - (2019), 9 pages. <https://doi.org/10.2312/CGVC.20191256>
- [13] David J Harris, Jonathan M Bird, Philip A Smart, Mark R Wilson, and Samuel J Vine. 2020. A framework for the testing and validation of simulated environments in experimentation and training. *Frontiers in Psychology* 11 (2020), 605.
- [14] R. Herpers, W. Heiden, M. Kutz, D. Scherfgen, U. Hartmann, J. Bongartz, and O. Schulzyk. 2008. FIVIS Bicycle Simulator: An Immersive Game Platform for Physical Activities. In *Proceedings of the 2008 Conference on Future Play: Research, Play, Share* (Toronto, Ontario, Canada) (Future Play '08). Association for Computing Machinery, New York, NY, USA, 244–247. <https://doi.org/10.1145/1496984.1497035>
- [15] Robert S Kennedy, Julie Drexler, and Robert C Kennedy. 2010. Research in visually induced motion sickness. *Applied ergonomics* 41, 4 (2010), 494–503.
- [16] Robert S Kennedy, Jennifer E Fowlkes, Kevin S Berbaum, and Michael G Lilienthal. 1992. Use of a motion sickness history questionnaire for prediction of simulator sickness. *Aviation, Space, and Environmental Medicine* 63, 7 (1992), 588–593.
- [17] Behrang Keshavarz and Heiko Hecht. 2011. Axis rotation and visually induced motion sickness: the role of combined roll, pitch, and yaw motion. *Aviat Space Environ Med* 82, 11 (Nov 2011), 1023–1029. <https://doi.org/10.3357/asm.3078.2011>
- [18] Jiwon Kim, Jihong Hwang, and Taezoon Park. 2020. Effect of motion cues on simulator sickness in a flight simulator. In *Virtual, Augmented and Mixed Reality. Design and Interaction: 12th International Conference, VAMR 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Copenhagen, Denmark, July 19–24, 2020, Proceedings, Part I* 22. Springer, Springer, Copenhagen, 493–506.
- [19] Jiwon Kim, Seong-Min Lee, Hungsun Son, and Taezoon Park. 2023. Effects of unlimited angular motion cue and cue discrepancy on simulator sickness. *Expert Systems with Applications* 213 (2023), 119196. <https://doi.org/10.1016/j.eswa.2022.119196>
- [20] J.D.G. Kooijman and A.L. Schwab. 2013. A review on bicycle and motorcycle rider control with a perspective on handling qualities. *Vehicle System Dynamics* 51, 11 (2013), 1722–1764. <https://doi.org/10.1080/00423114.2013.824990> arXiv:<https://doi.org/10.1080/00423114.2013.824990>
- [21] Natália Kováčová, Marco Grottolli, Francesco Celiberti, Yves Lemmens, Riender Happee, Marjan P. Hagenzieker, and Joost C.F. de Winter. 2020. Emergency

- braking at intersections: A motion-base motorcycle simulator study. *Applied Ergonomics* 82 (2020), 102970. <https://doi.org/10.1016/j.apergo.2019.102970>
- [22] Ernst Kruijff, Alexander Marquardt, Christina Trepkowski, Robert W. Lindeman, Andre Hinkenjann, Jens Maiero, and Bernhard E. Riecke. 2016. On Your Feet! Enhancing Vection in Learning-Based Interfaces through Multisensory Stimuli. In *Proceedings of the 2016 Symposium on Spatial User Interaction* (Tokyo, Japan) (SUI '16). Association for Computing Machinery, New York, NY, USA, 149–158. <https://doi.org/10.1145/2983310.2985759>
- [23] Marc Erich Latoschik and Carolin Wienrich. 2022. Congruence and Plausibility, Not Presence: Pivotal Conditions for XR Experiences and Effects, a Novel Approach. *Frontiers in Virtual Reality* 3 (2022). <https://doi.org/10.3389/frvir.2022.694433>
- [24] T. Maeda, H. Ando, and M. Sugimoto. 2005. Virtual acceleration with galvanic vestibular stimulation in a virtual reality environment. In *IEEE Proceedings. VR 2005. Virtual Reality*, 2005. IEEE, Bonn, Germany, 289–290. <https://doi.org/10.1109/VR.2005.1492799>
- [25] Andrii Matviienko, Swamy Ananthanarayan, Shadan Sadeghian Borojeni, Yannick Feld, Wilko Heuten, and Susanne Boll. 2018. Augmenting Bicycles and Helmets with Multimodal Warnings for Children. In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Barcelona, Spain) (MobileHCI '18). Association for Computing Machinery, New York, NY, USA, Article 15, 13 pages. <https://doi.org/10.1145/3229434.3229479>
- [26] Andrii Matviienko, Swamy Ananthanarayan, Stephen Brewster, Wilko Heuten, and Susanne Boll. 2019. Comparing Unimodal Lane Keeping Cues for Child Cyclists. In *Proceedings of the 18th International Conference on Mobile and Ubiquitous Multimedia* (Pisa, Italy) (MUM '19). Association for Computing Machinery, New York, NY, USA, Article 14, 11 pages. <https://doi.org/10.1145/3365610.3365632>
- [27] Andrii Matviienko, Swamy Ananthanarayan, Abdallah El Ali, Wilko Heuten, and Susanne Boll. 2019. NaviBike: Comparing Unimodal Navigation Cues for Child Cyclists. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300850>
- [28] Andrii Matviienko, Swamy Ananthanarayan, Raphael Kappes, Wilko Heuten, and Susanne Boll. 2020. Reminding Child Cyclists about Safety Gestures. In *Proceedings of the 9TH ACM International Symposium on Pervasive Displays* (Manchester, United Kingdom) (PerDis '20). Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3393712.3394120>
- [29] Andrii Matviienko, Florian Müller, Marcel Zickler, Lisa Alina Gasche, Julia Abels, Till Steinert, and Max Mühlhäuser. 2022. Reducing Virtual Reality Sickness for Cyclists in VR Bicycle Simulators. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 187, 14 pages. <https://doi.org/10.1145/3491102.3501959>
- [30] Justin Mittelstaedt, Jan Wacker, and Dirk Stelling. 2018. Effects of display type and motion control on cybersickness in a virtual bike simulator. *Displays* 51 (2018), 43–50. <https://doi.org/10.1016/j.displa.2018.01.002>
- [31] Takayuki Nozawa, Erwin Wu, and Hideki Koike. 2019. VR Ski Coach: Indoor Ski Training System Visualizing Difference from Leading Skier. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 1341–1342. <https://doi.org/10.1109/VR.2019.8797717>
- [32] Yi-Hao Peng, Carolyn Yu, Shi-Hong Liu, Chung-Wei Wang, Paul Taelle, Neng-Hao Yu, and Mike Y. Chen. 2020. *WalkingVibe: Reducing Virtual Reality Sickness and Improving Realism While Walking in VR Using Unobtrusive Head-Mounted Vibrotactile Feedback*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376847>
- [33] Sharif Razzaque. 2005. *Redirected walking*. The University of North Carolina at Chapel Hill.
- [34] Michael Rietzler, Teresa Hirzle, Jan Gugenheimer, Julian Frommel, Thomas Dreja, and Enrico Rukzio. 2018. VRSpinning: Exploring the Design Space of a 1D Rotation Platform to Increase the Perception of Self-Motion in VR. In *Proceedings of the 2018 Designing Interactive Systems Conference* (Hong Kong, China) (DIS '18). Association for Computing Machinery, New York, NY, USA, 99–108. <https://doi.org/10.1145/3196709.3196755>
- [35] Roy A. Ruddle, Ekaterina Volkova, and Heinrich H. Bühlhoff. 2011. Walking Improves Your Cognitive Map in Environments That Are Large-Scale and Large in Extent. *ACM Trans. Comput.-Hum. Interact.* 18, 2, Article 10 (July 2011), 20 pages. <https://doi.org/10.1145/1970378.1970384>
- [36] Barbara MA Schout, AJM Hendriks, Fedde Scheele, Bart LH Bemelmans, and AJJA Scherpier. 2010. Validation and implementation of surgical simulators: a critical review of present, past, and future. *Surgical endoscopy* 24 (2010), 536–546.
- [37] Richard Skarbez, Frederick P. Brooks, and Mary C. Whitton. 2021. Immersion and Coherence: Research Agenda and Early Results. *IEEE Transactions on Visualization and Computer Graphics* 27, 10 (2021), 3839–3850. <https://doi.org/10.1109/TVCG.2020.2983701>
- [38] Mel Slater. 2003. A note on presence terminology. *Presence connect* 3, 3 (2003), 1–5.
- [39] Mel Slater. 2009. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B: Biological Sciences* 364, 1535 (2009), 3549–3557.
- [40] Mel Slater. 2018. Immersion and the illusion of presence in virtual reality. *British Journal of Psychology* 109, 3 (2018), 431–433. <https://doi.org/10.1111/bjop.12305> arXiv:<https://arxiv.org/abs/1803.01230>
- [41] Ludwig Steindl. 2018. *Hybrid tracking technology for virtual rock climbing*. Ph.D. Dissertation. Wien.
- [42] Carlos Sun and Zhu Qing. 2018. Design and Construction of a Virtual Bicycle Simulator for Evaluating Sustainable Facilities Design. *Advances in Civil Engineering* 2018 (2018). <https://doi.org/10.1155/2018/5735820>
- [43] Léo Terziman, Maud Marchal, Franck Multon, Bruno Arnaldi, and Anatole Lecuyer. 2012. The King-Kong Effects: Improving sensation of walking in VR with visual and tactile vibrations at each step. In *2012 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, Costa Mesa, CA, 19–26. <https://doi.org/10.1109/3DUI.2012.6184179>
- [44] Sam Tregillus and Eelke Folmer. 2016. VR-STEP: Walking-in-Place Using Inertial Sensing for Hands Free Navigation in Mobile VR Environments. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 1250–1255. <https://doi.org/10.1145/2858036.2858084>
- [45] Huizhao Tu, Zhenfei Li, Hao Li, Ke Zhang, and Lijun Sun. 2015. Driving simulator fidelity and emergency driving behavior. *Transportation Research Record* 2518, 1 (2015), 113–121.
- [46] Daniela Ullmann, Julian Kreimeier, Timo Götzelmann, and Harald Kipke. 2020. BikeVR: A Virtual Reality Bicycle Simulator towards Sustainable Urban Space and Traffic Planning. In *Proceedings of the Conference on Mensch Und Computer* (Magdeburg, Germany) (MuC '20). Association for Computing Machinery, New York, NY, USA, 511–514. <https://doi.org/10.1145/3404983.3410417>
- [47] Martin Usoh, Kevin Arthur, Mary C. Whitton, Rui Bastos, Anthony Steed, Mel Slater, and Frederick P. Brooks. 1999. Walking > Walking-in-Place > Flying, in Virtual Environments. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '99)*. ACM Press/Addison-Wesley Publishing Co., USA, 359–364. <https://doi.org/10.1145/311535.311589>
- [48] Tamara von Sawitzky, Philipp Wintersberger, Andreas Löcken, Anna-Katharina Frison, and Andreas Riemer. 2020. Augmentation Concepts with HUDs for Cyclists to Improve Road Safety in Shared Spaces. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3334480.3383022>
- [49] Jia Wang and Robert W. Lindeman. 2012. Comparing isometric and elastic surfboard interfaces for leaning-based travel in 3D virtual environments. In *2012 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, Costa Mesa, CA, 31–38. <https://doi.org/10.1109/3DUI.2012.6184181>
- [50] Séamas Weech, Sophie Kenny, and Michael Barnett-Cowan. 2019. Presence and Cybersickness in Virtual Reality Are Negatively Related: A Review. *Frontiers in Psychology* 10 (2019). <https://doi.org/10.3389/fpsyg.2019.00158>
- [51] Séamas Weech, Jae Moon, and Nikolaus F. Troje. 2018. Influence of bone-conducted vibration on simulator sickness in virtual reality. *PLOS ONE* 13, 3 (03 2018), 1–21. <https://doi.org/10.1371/journal.pone.0194137>
- [52] Séamas Weech and Nikolaus F. Troje. 2017. Vection Latency Is Reduced by Bone-Conducted Vibration and Noisy Galvanic Vestibular Stimulation. *Multisensory Research* 30, 1 (2017), 65 – 90. <https://doi.org/10.1163/22134808-00002545>
- [53] Philipp Wintersberger, Andrii Matviienko, Andreas Schweidler, and Florian Michahelles. 2022. Development and Evaluation of a Motion-Based VR Bicycle Simulator. *Proc. ACM Hum.-Comput. Interact.* 6, MHCI, Article 210 (sep 2022), 19 pages. <https://doi.org/10.1145/3546745>